
Quadratic forms, symmetrization and contraction

In this chapter, we introduce a number of basic tools of high-dimensional probability: decoupling in Section 6.1, concentration of quadratic forms (the Hanson-Wright inequality) in Section 6.2, symmetrization in Section 6.4 and contraction in Section 6.7.

We illustrate these tools in a number of applications. In Section 6.3, we use the Hanson-Wright inequality to establish concentration for anisotropic random vectors (thus extending Theorem 3.1.1) and for the distances between random vectors and subspaces. In Section 6.5, we combine matrix Bernstein's inequality with symmetrization arguments to analyze the operator norm of a random matrix; we show that it is almost equivalent to the largest Euclidean norm of the rows and columns. We use this result in Section 6.6 for the problem of matrix completion, where one is shown a few randomly chosen entries of a given matrix and is asked to fill in the missing entries.

6.1 Decoupling

In the beginning of this book, we thoroughly studied independent random variables of the type

$$\sum_{i=1}^n a_i X_i \quad (6.1)$$

where $X_1, \dots, X_n$ are independent random variables and a_i are fixed coefficients. In this section, we study *quadratic forms* of the type

$$\sum_{i,j=1}^n a_{ij} X_i X_j = X^\top A X = \langle X, A X \rangle \quad (6.2)$$

where $A = (a_{ij})$ is an $n \times n$ matrix of coefficients, and $X = (X_1, \dots, X_n)$ is a random vector with independent coordinates. Such a quadratic form is called a *chaos* in probability theory.

Computing the expectation of a chaos is easy. For simplicity, let us assume that X_i have zero means and unit variances. Then

$$\mathbb{E} X^\top A X = \sum_{i,j=1}^n a_{ij} \mathbb{E} X_i X_j = \sum_{i=1}^n a_{ii} = \text{tr } A.$$

It is harder to establish a concentration of a chaos. The main difficulty is that the terms of the sum in (6.2) are not independent. This difficulty can be overcome by the *decoupling* technique, which we will study now.

The purpose of decoupling is to replace the quadratic form (6.2) with the *bilinear* form

$$\sum_{i,j=1}^n a_{ij} X_i X'_j = X^\top A X' = \langle X, A X' \rangle,$$

where $X' = (X'_1, \dots, X'_n)$ is a random vector which is independent of X yet has the same distribution as X . Such X' is called an *independent copy* of X . The point here is that the bilinear form is easier to analyze than the quadratic form, since it is linear in X . Indeed, if we condition on X' we may treat the bilinear form as a sum of independent random variables

$$\sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} X'_j \right) X_i = \sum_{i=1}^n c_i X_i$$

with fixed coefficients c_i , much like we treated the sums (6.1) before.

Theorem 6.1.1 (Decoupling). *Let A be an $n \times n$, diagonal-free matrix (i.e. the diagonal entries of A equal zero). Let $X = (X_1, \dots, X_n)$ be a random vector with independent mean zero coordinates X_i . Then, for every convex function $F : \mathbb{R} \rightarrow \mathbb{R}$, one has*

$$\mathbb{E} F(X^\top A X) \leq \mathbb{E} F(4X^\top A X') \quad (6.3)$$

where X' is an independent copy of X .

The proof will be based on the following observation.

Lemma 6.1.2. *Let Y and Z be independent random variables such that $\mathbb{E} Z = 0$. Then, for every convex function F , one has*

$$\mathbb{E} F(Y) \leq \mathbb{E} F(Y + Z).$$

Proof This is a simple consequence of Jensen's inequality. First let us fix an arbitrary $y \in \mathbb{R}$ and use $\mathbb{E} Z = 0$ to get

$$F(y) = F(y + \mathbb{E} Z) = F(\mathbb{E}[y + Z]) \leq \mathbb{E} F(y + Z).$$

Now choose $y = Y$ and take expectations of both sides to complete the proof. (To check if you understood this argument, find where the independence of Y and Z was used!) $\square$

Proof of Decoupling Theorem 6.1.1 Here is what our proof will look like in a nutshell. First, we replace the chaos $X^\top A X = \sum_{i,j} a_{ij} X_i X_j$ by the “partial chaos”

$$\sum_{(i,j) \in I \times I^c} a_{ij} X_i X_j$$

where the subset of indices $I \subset \{1, \dots, n\}$ will be chosen by random sampling.

The advantage of partial chaos is that the summation is done over disjoint sets for i and j . Thus one can automatically replace X_j by X'_j without changing the distribution. Finally, we complete the partial chaos to the full sum $X^\top AX' = \sum_{i,j} a_{ij} X_i X'_j$ using Lemma 6.1.2.

Now we pass to a detailed proof. To randomly select a subset of indices I , let us consider *selectors* $\delta_1, \dots, \delta_n \in \{0, 1\}$, which are independent Bernoulli random variables with $\mathbb{P}\{\delta_i = 0\} = \mathbb{P}\{\delta_i = 1\} = 1/2$. Define

$$I := \{i : \delta_i = 1\}.$$

Condition on X . Since by assumption $a_{ii} = 0$ and

$$\mathbb{E} \delta_i (1 - \delta_j) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad \text{for all } i \neq j,$$

we may express the chaos as

$$X^\top AX = \sum_{i \neq j} a_{ij} X_i X_j = 4 \mathbb{E}_\delta \sum_{i \neq j} \delta_i (1 - \delta_j) a_{ij} X_i X_j = 4 \mathbb{E}_I \sum_{(i,j) \in I \times I^c} a_{ij} X_i X_j.$$

(The subscripts δ and I are meant to remind us about the sources of randomness used in taking these conditional expectations. Since we fixed X , the conditional expectations are over the random selectors $\delta = (\delta_1, \dots, \delta_n)$, or equivalently, over the random set of indices I . We will continue to use similar notation later.)

Apply the function F to both sides and take expectation over X . Using Jensen's inequality and Fubini theorem, we obtain

$$\mathbb{E}_X F(X^\top AX) \leq \mathbb{E}_I \mathbb{E}_X F\left(4 \sum_{(i,j) \in I \times I^c} a_{ij} X_i X_j\right).$$

It follows that there exists a realization of a random subset I such that

$$\mathbb{E}_X F(X^\top AX) \leq \mathbb{E}_X F\left(4 \sum_{(i,j) \in I \times I^c} a_{ij} X_i X_j\right).$$

Fix such realization of I until the end of the proof (and drop the subscripts X in the expectation for convenience.) Since the random variables $(X_i)_{i \in I}$ are independent from $(X_j)_{j \in I^c}$, the distribution of the sum in the right side will not change if we replace X_j by X'_j . So we get

$$\mathbb{E} F(X^\top AX) \leq \mathbb{E} F\left(4 \sum_{(i,j) \in I \times I^c} a_{ij} X_i X'_j\right).$$

It remains to complete the sum in the right side to the sum over all pairs of indices. In other words, we want to show that

$$\mathbb{E} F\left(4 \sum_{(i,j) \in I \times I^c} a_{ij} X_i X'_j\right) \leq \mathbb{E} F\left(4 \sum_{(i,j) \in [n] \times [n]} a_{ij} X_i X'_j\right), \quad (6.4)$$

where we use the notation $[n] = \{1, \dots, n\}$. To do this, we decompose the sum

in the right side as

$$\sum_{(i,j) \in [n] \times [n]} a_{ij} X_i X'_j = Y + Z_1 + Z_2$$

where

$$Y = \sum_{(i,j) \in I \times I^c} a_{ij} X_i X'_j, \quad Z_1 = \sum_{(i,j) \in I \times I} a_{ij} X_i X'_j, \quad Z_2 = \sum_{(i,j) \in I^c \times [n]} a_{ij} X_i X'_j.$$

Condition on all random variables except $(X'_j)_{j \in I}$ and $(X_i)_{i \in I^c}$. This fixes Y , while Z_1 and Z_2 are random variables with zero conditional expectations (check!). Use Lemma 6.1.2 to conclude that the conditional expectation, which we denote $\mathbb{E}'$, satisfies

$$F(4Y) \leq \mathbb{E}' F(4Y + 4Z_1 + 4Z_2).$$

Finally, taking expectation of both sides over all other random variables, we conclude that

$$\mathbb{E} F(4Y) \leq \mathbb{E} F(4Y + 4Z_1 + 4Z_2).$$

This proves (6.4) and finishes the argument. $\square$

Remark 6.1.3. We actually proved a slightly stronger version of decoupling inequality, in which A needs not be diagonal-free. Thus, for any square matrix $A = (a_{ij})$ we showed that

$$\mathbb{E} F\left(\sum_{i,j: i \neq j} a_{ij} X_i X_j\right) \leq \mathbb{E} F\left(4 \sum_{i,j} a_{ij} X_i X'_j\right)$$

Exercise 6.1.4 (Decoupling in Hilbert spaces). $\clubsuit$ Prove the following generalization of Theorem 6.1.1. Let $A = (a_{ij})$ be an $n \times n$ matrix. Let $X_1, \dots, X_n$ be independent, mean zero random vectors in some Hilbert space. Show that for every convex function $F: \mathbb{R} \rightarrow \mathbb{R}$, one has

$$\mathbb{E} F\left(\sum_{i,j: i \neq j} a_{ij} \langle X_i, X_j \rangle\right) \leq \mathbb{E} F\left(4 \sum_{i,j} a_{ij} \langle X_i, X'_j \rangle\right),$$

where (X'_i) is an independent copy of (X_i) .

Exercise 6.1.5 (Decoupling in normed spaces). $\clubsuit\clubsuit$ Prove the following alternative generalization of Theorem 6.1.1. Let $(u_{ij})_{i,j=1}^n$ be fixed vectors in some normed space. Let $X_1, \dots, X_n$ be independent, mean zero random variables. Show that, for every convex and increasing function F , one has

$$\mathbb{E} F\left(\left\|\sum_{i,j: i \neq j} X_i X_j u_{ij}\right\|\right) \leq \mathbb{E} F\left(4 \left\|\sum_{i,j} X_i X'_j u_{ij}\right\|\right).$$

where (X'_i) is an independent copy of (X_i) .

6.2 Hanson-Wright Inequality

We now prove a general concentration inequality for a chaos. It can be viewed as a chaos version of Bernstein's inequality.

Theorem 6.2.1 (Hanson-Wright inequality). *Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, sub-gaussian coordinates. Let A be an $n \times n$ matrix. Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ |X^\top A X - \mathbb{E} X^\top A X| \geq t \right\} \leq 2 \exp \left[-c \min \left(\frac{t^2}{K^4 \|A\|_F^2}, \frac{t}{K^2 \|A\|} \right) \right],$$

where $K = \max_i \|X_i\|_{\psi_2}$.

Like many times before, our proof of Hanson-Wright inequality will be based on bounding the moment generating function of $X^\top A X$. We use decoupling to replace this chaos by $X^\top A X'$. Next, we bound the MGF of the decoupled chaos in the easier, Gaussian case where $X \sim N(0, I_n)$. Finally, we extend the bound to general sub-gaussian distributions using a replacement trick.

Lemma 6.2.2 (MGF of Gaussian chaos). *Let $X, X' \sim N(0, I_n)$ be independent and let $A = (a_{ij})$ be an $n \times n$ matrix. Then*

$$\mathbb{E} \exp(\lambda X^\top A X') \leq \exp(C \lambda^2 \|A\|_F^2)$$

for all λ satisfying $|\lambda| \leq c/\|A\|$.

Proof First let us use rotation invariance to reduce to the case where matrix A is diagonal. Expressing A through its singular value decomposition

$$A = \sum_i s_i u_i v_i^\top,$$

we can write

$$X^\top A X' = \sum_i s_i \langle u_i, X \rangle \langle v_i, X' \rangle.$$

By rotation invariance of the normal distribution, $g := (\langle u_i, X \rangle)_{i=1}^n$ and $g' := (\langle v_i, X' \rangle)_{i=1}^n$ are independent standard normal random vectors in $\mathbb{R}^n$ (recall Exercise 3.3.3). In other words, we represented the chaos as

$$X^\top A X' = \sum_i s_i g_i g'_i$$

where $g, g' \sim N(0, I_n)$ are independent and s_i are the singular values of A .

This is a sum of independent random variables, which is easy to handle. Indeed, independence gives

$$\mathbb{E} \exp(\lambda X^\top A X') = \prod_i \mathbb{E} \exp(\lambda s_i g_i g'_i). \quad (6.5)$$

Now, for each i , we have

$$\mathbb{E} \exp(\lambda s_i g_i g'_i) = \mathbb{E} \exp(\lambda^2 s_i^2 g_i^2 / 2) \leq \exp(C \lambda^2 s_i^2) \quad \text{provided that } \lambda^2 s_i^2 \leq c.$$

To get the first identity here, condition on g_i and use the formula (2.12) for the MGF of the normal random variable g'_i . At the second step, we used part c of Proposition 2.7.1 for the sub-exponential random variable g_i^2 .

Substituting this bound into (6.5), we obtain

$$\mathbb{E} \exp(\lambda X^\top A X') \leq \exp\left(C\lambda^2 \sum_i s_i^2\right) \quad \text{provided that } \lambda^2 \leq \frac{c}{\max_i s_i^2}.$$

It remains to recall that s_i are the singular values of A , so $\sum_i s_i^2 = \|A\|_F^2$ and $\max_i s_i = \|A\|$. The lemma is proved. $\square$

To extend Lemma 6.2.2 to general distributions, we use a replacement trick to compare the MGF's of general and Gaussian chaoses.

Lemma 6.2.3 (Comparison). *Consider independent, mean zero, sub-gaussian random vectors X, X' in $\mathbb{R}^n$ with $\|X\|_{\psi_2} \leq K$ and $\|X'\|_{\psi_2} \leq K$. Consider also independent random vectors $g, g' \sim N(0, I_n)$. Let A be an $n \times n$ matrix. Then*

$$\mathbb{E} \exp(\lambda X^\top A X') \leq \mathbb{E} \exp(CK^2 \lambda g^\top A g')$$

for any $\lambda \in \mathbb{R}$.

Proof Condition on X' and take expectation over X , which we denote $\mathbb{E}_X$. Then the random variable $X^\top A X' = \langle X, A X' \rangle$ is (conditionally) sub-gaussian, and its sub-gaussian norm¹ is bounded by $K\|A X'\|_2$. Then the bound (2.16) on the MGF of sub-gaussian random variables gives

$$\mathbb{E}_X \exp(\lambda X^\top A X') \leq \exp(C\lambda^2 K^2 \|A X'\|_2^2), \quad \lambda \in \mathbb{R}. \quad (6.6)$$

Compare this to the formula (2.12) for the MGF of the normal distribution. Applied to the normal random variable $g^\top A X' = \langle g, A X' \rangle$ (still conditionally on X'), it gives

$$\mathbb{E}_g \exp(\mu g^\top A X') = \exp(\mu^2 \|A X'\|_2^2 / 2), \quad \mu \in \mathbb{R}. \quad (6.7)$$

Choosing $\mu = \sqrt{2C}K\lambda$, we match the right sides of (6.6) and (6.7) and thus get

$$\mathbb{E}_X \exp(\lambda X^\top A X') \leq \mathbb{E}_g \exp(\sqrt{2C}K\lambda g^\top A X').$$

Taking expectation over X' of both sides, we see that we have successfully replaced X by g in the chaos, and we paid a factor of $\sqrt{2C}K$. Doing a similar argument again, this time for X' , we can further replace X' with g' and pay an extra factor of $\sqrt{2C}K$. (Exercise 6.2.4 below asks you to carefully write the details of this step.) The proof of lemma is complete. $\square$

Exercise 6.2.4 (Comparison).  Complete the proof of Lemma 6.2.3. Replace X' by g' ; write all details carefully.

¹ Recall Definition 3.4.1.

Proof of Theorem 6.2.1 Without loss of generality, we may assume that $K = 1$. (Why?) As usual, it is enough to bound the one-sided tail

$$p := \mathbb{P} \left\{ X^\top A X - \mathbb{E} X^\top A X \geq t \right\}.$$

Indeed, once we have a bound on this upper tail, a similar bound will hold for the lower tail as well (since one can replace A with $-A$). By combining the two tails, we would complete the proof.

In terms of the entries of $A = (a_{ij})_{i,j=1}^n$, we have

$$X^\top A X = \sum_{i,j} a_{ij} X_i X_j \quad \text{and} \quad \mathbb{E} X^\top A X = \sum_i a_{ii} \mathbb{E} X_i^2,$$

where we used the mean zero assumption and independence. So we can express the deviation as

$$X^\top A X - \mathbb{E} X^\top A X = \sum_i a_{ii} (X_i^2 - \mathbb{E} X_i^2) + \sum_{i,j: i \neq j} a_{ij} X_i X_j.$$

The problem reduces to estimating the diagonal and off-diagonal sums:

$$p \leq \mathbb{P} \left\{ \sum_i a_{ii} (X_i^2 - \mathbb{E} X_i^2) \geq t/2 \right\} + \mathbb{P} \left\{ \sum_{i,j: i \neq j} a_{ij} X_i X_j \geq t/2 \right\} =: p_1 + p_2.$$

Step 1: diagonal sum. Since X_i are independent, sub-gaussian random variables, $X_i^2 - \mathbb{E} X_i^2$ are independent, mean-zero, sub-exponential random variables, and

$$\|X_i^2 - \mathbb{E} X_i^2\|_{\psi_1} \lesssim \|X_i^2\|_{\psi_1} \lesssim \|X_i\|_{\psi_2}^2 \lesssim 1.$$

(This follows from the Centering Exercise 2.7.10 and Lemma 2.7.6.) Then Bernstein's inequality (Theorem 2.8.2) gives

$$p_1 \leq \exp \left[-c \min \left(\frac{t^2}{\sum_i a_{ii}^2}, \frac{t}{\max_i |a_{ii}|} \right) \right] \leq \exp \left[-c \min \left(\frac{t^2}{\|A\|_F^2}, \frac{t}{\|A\|} \right) \right].$$

Step 2: off-diagonal sum. It remains to bound the off-diagonal sum

$$S := \sum_{i,j: i \neq j} a_{ij} X_i X_j.$$

Let $\lambda > 0$ be a parameter whose value we will determine later. By Markov's inequality, we have

$$p_2 = \mathbb{P} \{ S \geq t/2 \} = \mathbb{P} \{ \lambda S \geq \lambda t/2 \} \leq \exp(-\lambda t/2) \mathbb{E} \exp(\lambda S). \quad (6.8)$$

Now,

$$\begin{aligned} \mathbb{E} \exp(\lambda S) &\leq \mathbb{E} \exp(4\lambda X^\top A X') \quad (\text{by decoupling -- see Remark 6.1.3}) \\ &\leq \mathbb{E} \exp(C_1 \lambda g^\top A g') \quad (\text{by Comparison Lemma 6.2.3}) \\ &\leq \exp(C \lambda^2 \|A\|_F^2) \quad (\text{by Lemma 6.2.2 for Gaussian chaos}), \end{aligned}$$

provided that $|\lambda| \leq c/\|A\|$. Putting this bound into (6.8), we obtain

$$p_2 \leq \exp\left(-\lambda t/2 + C\lambda^2\|A\|_F^2\right).$$

Optimizing over $0 \leq \lambda \leq c/\|A\|$, we conclude that

$$p_2 \leq \exp\left[-c \min\left(\frac{t^2}{\|A\|_F^2}, \frac{t}{\|A\|}\right)\right].$$

(Check!)

Summarizing, we obtained the desired bounds for the probabilities of diagonal deviation p_1 and off-diagonal deviation p_2 . Putting them together, we complete the proof of Theorem 6.2.1.²

□

Exercise 6.2.5. ☛☛☛ Give an alternative proof of Hanson-Wright inequality for normal distributions, without separating the diagonal part or decoupling.

Hint: Use the singular value decomposition for A and rotation invariance of $X \sim N(0, I_n)$ to simplify and control the quadratic form $X^T A X$.

Exercise 6.2.6. ☛☛☛ Consider a mean zero, sub-gaussian random vector X in $\mathbb{R}^n$ with $\|X\|_{\psi_2} \leq K$. Let B be an $m \times n$ matrix. Show that

$$\mathbb{E} \exp(\lambda^2 \|BX\|_2^2) \leq \exp(CK^2 \lambda^2 \|B\|_F^2) \quad \text{provided } |\lambda| \leq \frac{c}{K\|B\|}.$$

To prove this bound, replace X with a Gaussian random vector $g \sim N(0, I_m)$ along the following lines:

(a) Prove the comparison inequality

$$\mathbb{E} \exp(\lambda^2 \|BX\|_2^2) \leq \mathbb{E} \exp(CK^2 \lambda^2 \|B^T g\|_2^2)$$

for every $\lambda \in \mathbb{R}$. **Hint:** Argue like in the proof of Comparison Lemma 6.2.3.

(b) Check that

$$\mathbb{E} \exp(\lambda^2 \|B^T g\|_2^2) \leq \exp(C\lambda^2 \|B\|_F^2)$$

provided that $|\lambda| \leq c/\|B\|$. **Hint:** Argue like in the proof of Lemma 6.2.2.

Exercise 6.2.7 (Higher-dimensional Hanson-Wright inequality). ☛☛☛ Let $X_1, \dots, X_n$ be independent, mean zero, sub-gaussian random vectors in $\mathbb{R}^d$. Let $A = (a_{ij})$ be an $n \times n$ matrix. Prove that for every $t \geq 0$, we have

$$\mathbb{P}\left\{\left|\sum_{i,j: i \neq j}^n a_{ij} \langle X_i, X_j \rangle\right| \geq t\right\} \leq 2 \exp\left[-c \min\left(\frac{t^2}{K^4 d \|A\|_F^2}, \frac{t}{K^2 \|A\|}\right)\right]$$

where $K = \max_i \|X_i\|_{\psi_2}$.

Hint: The quadratic form in question can be represented as $X^T A X$ like before, but now X is a $d \times n$ random

² The reader might notice that our argument gives a factor 4 instead of 2 in front of the probability bound in Theorem 6.2.1 – this comes from adding the bounds for the upper and lower tails.

However, one can replace 4 by 2 by lowering the constant c in the exponent appropriately. (How?)

matrix with columns X_i . Redo the computation for the MGF when X is Gaussian (Lemma 6.2.2) and the Comparison Lemma 6.2.3.

6.3 Concentration of anisotropic random vectors

As a consequence of Hanson-Wright inequality, we now obtain concentration for *anisotropic* random vectors, which have the form BX , where B is a fixed matrix and X is an isotropic random vector.

Exercise 6.3.1. ☛ Let B be an $m \times n$ matrix and X is an isotropic random vector in $\mathbb{R}^n$. Check that

$$\mathbb{E} \|BX\|_2^2 = \|B\|_F^2.$$

Theorem 6.3.2 (Concentration of random vectors). *Let B be an $m \times n$ matrix, and let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, unit variance, sub-gaussian coordinates. Then*

$$\left\| \|BX\|_2 - \|B\|_F \right\|_{\psi_2} \leq CK^2 \|B\|,$$

where $K = \max_i \|X_i\|_{\psi_2}$.

An important partial case of this theorem when $B = I_n$. In this case, the inequality we obtain is

$$\left\| \|X\|_2 - \sqrt{n} \right\|_{\psi_2} \leq CK^2,$$

which we proved in Theorem 3.1.1.

Proof of Theorem 6.3.2. For simplicity, we assume that $K \geq 1$. (Argue that you can make this assumption.) We apply Hanson-Wright inequality (Theorem 6.2.1) for the matrix $A := B^\top B$. Let us express the main terms appearing in Hanson-Wright inequality in terms of B . We have

$$X^\top AX = \|BX\|_2^2, \quad \mathbb{E} X^\top AX = \|B\|_F^2$$

and

$$\|A\| = \|B\|^2, \quad \|A\|_F = \|B^\top B\|_F \leq \|B^\top\| \|B\|_F = \|B\| \|B\|_F.$$

(You will be asked to check the inequality in Exercise 6.3.3.) Thus, we have for every $u \geq 0$ that

$$\mathbb{P} \left\{ \left| \|BX\|_2^2 - \|B\|_F^2 \right| \geq u \right\} \leq 2 \exp \left[- \frac{c}{K^4} \min \left(\frac{u^2}{\|B\|^2 \|B\|_F^2}, \frac{u}{\|B\|^2} \right) \right].$$

(Here we used that $K^4 \geq K^2$ since we assumed that $K \geq 1$.)

Substitute the value $u = \varepsilon \|B\|_F^2$ for $\varepsilon \geq 0$ and obtain

$$\mathbb{P} \left\{ \left| \|BX\|_2^2 - \|B\|_F^2 \right| \geq \varepsilon \|B\|_F^2 \right\} \leq 2 \exp \left[- c \min(\varepsilon^2, \varepsilon) \frac{\|B\|_F^2}{K^4 \|B\|^2} \right].$$

This is a good concentration inequality for $\|BX\|_2^2$, from which we are going to deduce a concentration inequality for $\|BX\|_2$. Denote $\delta^2 = \min(\varepsilon^2, \varepsilon)$, or equivalently set $\varepsilon = \max(\delta, \delta^2)$. Observe that the following implication holds:

$$\text{If } \left| \|BX\|_2 - \|B\|_F \right| \geq \delta \|B\|_F \quad \text{then} \quad \left| \|BX\|_2^2 - \|B\|_F^2 \right| \geq \varepsilon \|B\|_F^2.$$

(Check it! – This is the same elementary inequality as (3.2), once we divide through by $\|B\|_F^2$.) Thus we get

$$\mathbb{P} \left\{ \left| \|BX\|_2 - \|B\|_F \right| \geq \delta \|B\|_F \right\} \leq 2 \exp \left(-c\delta^2 \frac{\|B\|_F^2}{K^4 \|B\|^2} \right).$$

Changing variables to $t = \delta \|B\|_F$, we obtain

$$\mathbb{P} \left\{ \left| \|BX\|_2 - \|B\|_F \right| > t \right\} \leq 2 \exp \left(-\frac{ct^2}{K^4 \|B\|^2} \right).$$

Since this inequality holds for all $t \geq 0$, the conclusion of the theorem follows from the definition of sub-gaussian distributions. $\square$

Exercise 6.3.3.  Let D be a $k \times m$ matrix and B be an $m \times n$ matrix. Prove that

$$\|DB\|_F \leq \|D\| \|B\|_F.$$

Exercise 6.3.4 (Distance to a subspace).  Let E be a subspace of $\mathbb{R}^n$ of dimension d . Consider a random vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ with independent, mean zero, unit variance, sub-gaussian coordinates.

(a) Check that

$$(\mathbb{E} \text{dist}(X, E)^2)^{1/2} = \sqrt{n-d}.$$

(b) Prove that for any $t \geq 0$, the distance nicely concentrates:

$$\mathbb{P} \left\{ \left| d(X, E) - \sqrt{n-d} \right| > t \right\} \leq 2 \exp(-ct^2/K^4),$$

where $K = \max_i \|X_i\|_{\psi_2}$.

Let us prove a weaker version of Theorem 6.3.2 without assuming independence of the coordinates of X :

Exercise 6.3.5 (Tails of sub-gaussian random vectors).  Let B be an $m \times n$ matrix, and let X be a mean zero, sub-gaussian random vector in $\mathbb{R}^n$ with $\|X\|_{\psi_2} \leq K$. Prove that for any $t \geq 0$, we have

$$\mathbb{P} \left\{ \|BX\|_2 \geq CK \|B\|_F + t \right\} \leq \exp \left(-\frac{ct^2}{K^2 \|B\|^2} \right).$$

Hint: Use the bound on the MGF we proved Exercise 6.2.6.

The following exercise explains why the concentration inequality *must* be weaker than in Theorem 3.1.1 if we do not assume independence of coordinates of X .

Exercise 6.3.6. ☕ Show that there exists a mean zero, isotropic, and sub-gaussian random vector X in $\mathbb{R}^n$ such that

$$\mathbb{P}\{\|X\|_2 = 0\} = \mathbb{P}\{\|X\|_2 \geq 1.4\sqrt{n}\} = \frac{1}{2}.$$

In other words, $\|X\|_2$ does not concentrate near $\sqrt{n}$.

6.4 Symmetrization

A random variable X is *symmetric* if X and $-X$ have the same distribution. A simplest example of a symmetric random variable is *symmetric Bernoulli*, which takes values -1 and 1 with probabilities $1/2$ each:

$$\mathbb{P}\{\xi = 1\} = \mathbb{P}\{\xi = -1\} = \frac{1}{2}.$$

A normal, mean zero random variable $X \sim N(0, \sigma^2)$ is also symmetric, while Poisson or exponential random variables are not.

In this section we develop the simple and useful technique of *symmetrization*. It allows one to reduce problems about arbitrary distributions to symmetric distributions, and in some cases even to the symmetric Bernoulli distribution.

Exercise 6.4.1 (Constructing symmetric distributions). ☕ Let X be a random variable and ξ be an independent symmetric Bernoulli random variable.

- (a) Check that ξX and $\xi|X|$ are symmetric random variables, and they have the same distribution.
- (b) If X is symmetric, show that the distribution of ξX and $\xi|X|$ is the same as of X .
- (c) Let X' be an independent copy of X . Check that $X - X'$ is symmetric.

Throughout this section, we denote by

$$\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots$$

a sequence of independent symmetric Bernoulli random variables. We assume that they are (jointly) independent not only of each other, but also of any other random variables in question.

Lemma 6.4.2 (Symmetrization). *Let $X_1, \dots, X_N$ be independent, mean zero random vectors in a normed space. Then*

$$\frac{1}{2} \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|.$$

The purpose of this lemma is to let us replace general random variables X_i by the symmetric random variables $\varepsilon_i X_i$.

Proof **Upper bound.** Let (X'_i) be an independent copy of the random vectors (X_i) . Since $\sum_i X'_i$ has zero mean, we have

$$p := \mathbb{E} \left\| \sum_i X_i \right\| \leq \mathbb{E} \left\| \sum_i X_i - \sum_i X'_i \right\| = \mathbb{E} \left\| \sum_i (X_i - X'_i) \right\|.$$

The inequality here is an application of the following version of Lemma 6.1.2 for independent random vectors Y and Z :

$$\text{if } \mathbb{E} Z = 0 \quad \text{then } \mathbb{E} \|Y\| \leq \mathbb{E} \|Y + Z\|. \quad (6.9)$$

(Check it!)

Next, since $(X_i - X'_i)$ are symmetric random vectors, they have the same distribution as $\varepsilon_i(X_i - X'_i)$ (see Exercise 6.4.1). Then

$$\begin{aligned} p &\leq \mathbb{E} \left\| \sum_i \varepsilon_i (X_i - X'_i) \right\| \\ &\leq \mathbb{E} \left\| \sum_i \varepsilon_i X_i \right\| + \mathbb{E} \left\| \sum_i \varepsilon_i X'_i \right\| \quad (\text{by triangle inequality}) \\ &= 2 \mathbb{E} \left\| \sum_i \varepsilon_i X_i \right\| \quad (\text{since the two terms are identically distributed}). \end{aligned}$$

Lower bound. The argument here is similar:

$$\begin{aligned} \mathbb{E} \left\| \sum_i \varepsilon_i X_i \right\| &\leq \mathbb{E} \left\| \sum_i \varepsilon_i (X_i - X'_i) \right\| \quad (\text{condition on } (\varepsilon_i) \text{ and use (6.9)}) \\ &= \mathbb{E} \left\| \sum_i (X_i - X'_i) \right\| \quad (\text{the distribution is the same}) \\ &\leq \mathbb{E} \left\| \sum_i X_i \right\| + \mathbb{E} \left\| \sum_i X'_i \right\| \quad (\text{by triangle inequality}) \\ &\leq 2 \mathbb{E} \left\| \sum_i X_i \right\| \quad (\text{by identical distribution}). \end{aligned}$$

This completes the proof of the symmetrization lemma. $\square$

Exercise 6.4.3. ☕☕ Where in this argument did we use the independence of the random variables X_i ? Is mean zero assumption needed for both upper and lower bounds?

Exercise 6.4.4 (Removing the mean zero assumption). ☕☕

- (a) Prove the following generalization of Symmetrization Lemma 6.4.2 for random vectors X_i that do not necessarily have zero means:

$$\mathbb{E} \left\| \sum_{i=1}^N X_i - \sum_{i=1}^N \mathbb{E} X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|.$$

- (b) Argue that there can not be any non-trivial reverse inequality.

Exercise 6.4.5. ☞ Prove the following generalization of Symmetrization Lemma 6.4.2. Let $F : \mathbb{R}_+ \rightarrow \mathbb{R}$ be an increasing, convex function. Show that the same inequalities in Lemma 6.4.2 hold if the norm $\|\cdot\|$ is replaced with $F(\|\cdot\|)$, namely

$$\mathbb{E} F\left(\frac{1}{2} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right) \leq \mathbb{E} F\left(\left\| \sum_{i=1}^N X_i \right\| \right) \leq \mathbb{E} F\left(2 \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right).$$

Exercise 6.4.6. ☞☞ Let $X_1, \dots, X_N$ be independent, mean zero random variables. Show that their sum $\sum_i X_i$ is sub-gaussian if and only if $\sum_i \varepsilon_i X_i$ is sub-gaussian, and

$$c \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|_{\psi_2} \leq \left\| \sum_{i=1}^N X_i \right\|_{\psi_2} \leq C \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|_{\psi_2}.$$

Hint: Use the result of Exercise 6.4.5 with $F(x) = \exp(\lambda x)$ to bound the moment generating function, or with $F(x) = \exp(cx^2)$.

6.5 Random matrices with non-i.i.d. entries

A typical usage of symmetrization technique consists of two steps. First, general random variables X_i are replaced by symmetric random variables $\varepsilon_i X_i$. Next, one conditions on X_i , which leaves the entire randomness with ε_i . This reduces the problem to *symmetric Bernoulli* random variables ε_i , which are often simpler to deal with. We illustrate this technique by proving a general bound on the norms of random matrices with independent but not identically distributed entries.

Theorem 6.5.1 (Norms of random matrices with non-i.i.d. entries). *Let A be an $n \times n$ symmetric random matrix whose entries on and above the diagonal are independent, mean zero random variables. Then*

$$\mathbb{E} \|A\| \leq C \sqrt{\log n} \cdot \mathbb{E} \max_i \|A_i\|_2,$$

where A_i denote the rows of A .

Before we pass to the proof of this theorem, let us note that it is sharp up to the logarithmic factor. Indeed, since the operator norm of any matrix is bounded below by the Euclidean norms of the rows (why?), we trivially have

$$\mathbb{E} \|A\| \geq \mathbb{E} \max_i \|A_i\|_2.$$

Note also that unlike all results we have seen before, Theorem 6.5.1 does not require *any moment assumptions* on the entries of A .

Proof of Theorem 6.5.1 Our argument will be based on a combination of symmetrization with matrix Khintchine's inequality (Exercise 5.4.13).

First decompose A into a sum of independent, mean zero, symmetric random matrices Z_{ij} , each of which contains a pair of symmetric entries of A (or one

diagonal entry). Precisely, we have

$$A = \sum_{i \leq j} Z_{ij}, \quad \text{where} \quad Z_{ij} := \begin{cases} A_{ij}(e_i e_j^\top + e_j e_i^\top), & i < j \\ A_{ii} e_i e_i^\top, & i = j \end{cases}$$

and where (e_i) denotes the canonical basis of $\mathbb{R}^n$.

Apply Symmetrization Lemma 6.4.2, which gives

$$\mathbb{E} \|A\| = \mathbb{E} \left\| \sum_{i \leq j} Z_{ij} \right\| \leq 2 \mathbb{E} \left\| \sum_{i \leq j} \varepsilon_{ij} Z_{ij} \right\|, \quad (6.10)$$

where (ε_{ij}) are independent symmetric Bernoulli random variables.

Condition on (Z_{ij}) and apply matrix Khintchine's inequality (Exercise 5.4.13); then take expectation with respect to (Z_{ij}) . This gives

$$\mathbb{E} \left\| \sum_{i \leq j} \varepsilon_{ij} Z_{ij} \right\| \leq C \sqrt{\log n} \mathbb{E} \left(\left\| \sum_{i \leq j} Z_{ij}^2 \right\| \right)^{1/2} \quad (6.11)$$

Now, a quick check verifies that each Z_{ij}^2 is a diagonal matrix; more precisely

$$Z_{ij}^2 = \begin{cases} A_{ij}^2 (e_i e_i^\top + e_j e_j^\top), & i < j \\ A_{ii}^2 e_i e_i^\top, & i = j. \end{cases}$$

Summing up, we get

$$\sum_{i \leq j} Z_{ij}^2 = \sum_{i=1}^n \left(\sum_{j=1}^n A_{ij}^2 \right) e_i e_i^\top = \sum_{i=1}^n \|A_i\|_2^2 e_i e_i^\top.$$

(Check the first identity carefully!) In other words, $\sum_{i \leq j} Z_{ij}^2$ is a diagonal matrix, and its diagonal entries equal $\|A_i\|_2^2$. The operator norm of a diagonal matrix is the maximal absolute value of its entries (why?), thus

$$\left\| \sum_{i \leq j} Z_{ij}^2 \right\| = \max_i \|A_i\|_2^2.$$

Substitute this into (6.11) and then into (6.10) and complete the proof. $\square$

In the following exercise, we derive a version of Theorem 6.5.1 for non-symmetric, rectangular matrices using the so-called “Hermitization trick”.

Exercise 6.5.2 (Rectangular matrices).  Let A be an $m \times n$ random matrix whose entries are independent, mean zero random variables. Show that

$$\mathbb{E} \|A\| \leq C \sqrt{\log(m+n)} \left(\mathbb{E} \max_i \|A_i\|_2 + \mathbb{E} \max_j \|A^j\|_2 \right)$$

where A_i and A^j denote the rows and columns of A , respectively.

Hint: Apply Theorem 6.5.1 for the $(m+n) \times (m+n)$ symmetric random matrix $\begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix}$.

Exercise 6.5.3 (Sharpness). 🍄 Show that the result of Exercise 6.5.2 is sharp up to the logarithmic factor, i.e. one always has

$$\mathbb{E} \|A\| \geq c \left(\mathbb{E} \max_i \|A_i\|_2 + \mathbb{E} \max_j \|A^j\|_2 \right).$$

Exercise 6.5.4 (Sharpness). 🍄🍄🍄 Show that the logarithmic factor in Theorem 6.5.1 can not be completely removed in general: construct a random matrix A satisfying the assumptions of the theorem and for which

$$\mathbb{E} \|A\| \geq c \log^{1/4}(n) \cdot \mathbb{E} \max_i \|A_i\|_2.$$

Hint: Let A be a block-diagonal matrix with independent n/k blocks; let each block be a $k \times k$ symmetric random matrix with independent symmetric Bernoulli random variables on and above diagonal. Condition on the event that there exists a block whose all entries equal 1. At the end of the argument, choose the value of k appropriately.

6.6 Application: matrix completion

A remarkable application of the methods we have studied is to the problem of matrix completion. Suppose we are shown a few entries of a matrix; can we guess the other entries? We obviously can not unless we know something else about the matrix. In this section we show that if the matrix has *low rank* then matrix completion is possible.

To describe the problem mathematically, consider a fixed $n \times n$ matrix X with

$$\text{rank}(X) = r$$

where $r \ll n$. Suppose we are shown a few *randomly chosen entries* of X . Each entry X_{ij} is revealed to us independently with some probability $p \in (0, 1)$ and is hidden from us with probability $1 - p$. In other words, assume that we are shown the $n \times n$ matrix Y whose entries are

$$Y_{ij} := \delta_{ij} X_{ij} \quad \text{where} \quad \delta_{ij} \sim \text{Ber}(p) \text{ are independent.}$$

These δ_{ij} are *selectors* – Bernoulli random variables that indicate whether an entry is revealed to us or not (in the latter case, it is replaced with zero). If

$$p = \frac{m}{n^2} \tag{6.12}$$

then we are *shown m entries of X on average*.

How can we infer X from Y ? Although X has small rank r by assumption, Y may not have small rank. (Why?) It is thus natural to enforce small rank by choosing a *best rank r approximation* to Y . (Recall the notion of best rank k approximation by looking back at Section 4.1.4.) The result, properly scaled, will be a good approximation to X :

Theorem 6.6.1 (Matrix completion). *Let $\hat{X}$ be a best rank r approximation to $p^{-1}Y$. Then*

$$\mathbb{E} \frac{1}{n} \|\hat{X} - X\|_F \leq C \sqrt{\frac{rn \log n}{m}} \|X\|_\infty,$$

as long as $m \geq n \log n$. Here $\|X\|_\infty = \max_{i,j} |X_{ij}|$ is the maximum magnitude of the entries of X .

Before we pass to the proof, let us pause quickly to note that Theorem 6.6.1 bounds the recovery error

$$\frac{1}{n} \|\hat{X} - X\|_F = \left(\frac{1}{n^2} \sum_{i,j=1}^n |\hat{X}_{ij} - X_{ij}|^2 \right)^{1/2}.$$

This is simply the average error per entry (in the L^2 sense). If we choose the average number of observed entries m so that

$$m \geq C' r n \log n$$

with large constant C' , then Theorem 6.6.1 guarantees that the average error is much smaller than $\|X\|_\infty$.

To summarize, *matrix completion is possible if the number of observed entries exceeds rn by a logarithmic margin*. In this case, the expected average error per entry is much smaller than the maximal magnitude of an entry. Thus, for low rank matrices, matrix completion is possible with few observed entries.

Proof We first bound the recovery error in the operator norm, and then pass to the Frobenius norm using the low rank assumption.

Step 1: bounding the error in the operator norm. Using the triangle inequality, let us split the error as follows:

$$\|\hat{X} - X\| \leq \|\hat{X} - p^{-1}Y\| + \|p^{-1}Y - X\|.$$

Since we have chosen $\hat{X}$ as a best rank r approximation to $p^{-1}Y$, the second summand dominates, i.e. $\|\hat{X} - p^{-1}Y\| \leq \|p^{-1}Y - X\|$, so we have

$$\|\hat{X} - X\| \leq 2\|p^{-1}Y - X\| = \frac{2}{p}\|Y - pX\|. \quad (6.13)$$

Note that the matrix $\hat{X}$, which would be hard to handle, has disappeared from the bound. Instead, $Y - pX$ is a matrix that is easy to understand. Its entries

$$(Y - pX)_{ij} = (\delta_{ij} - p)X_{ij}$$

are independent and mean zero random variables. So we can apply the result of Exercise 6.5.2, which gives

$$\mathbb{E} \|Y - pX\| \leq C\sqrt{\log n} \left(\mathbb{E} \max_{i \in [n]} \|(Y - pX)_i\|_2 + \mathbb{E} \max_{j \in [n]} \|(Y - pX)^j\|_2 \right). \quad (6.14)$$

To bound the norms of the rows and columns of $Y - pX$, we can express them as

$$\|(Y - pX)_i\|_2^2 = \sum_{j=1}^n (\delta_{ij} - p)^2 X_{ij}^2 \leq \sum_{j=1}^n (\delta_{ij} - p)^2 \cdot \|X\|_\infty^2,$$

and similarly for columns. These sums of independent random variables can be easily bounded using Bernstein's (or Chernoff's) inequality, which yields

$$\mathbb{E} \max_{i \in [n]} \sum_{j=1}^n (\delta_{ij} - p)^2 \leq Cpn.$$

(We do this calculation in Exercise 6.6.2.) Combining with a similar bound for the columns and substituting into (6.14), we obtain

$$\mathbb{E} \|Y - pX\| \lesssim \sqrt{pn \log n} \|X\|_\infty.$$

Then, by (6.13), we get

$$\mathbb{E} \|\hat{X} - X\| \lesssim \sqrt{\frac{n \log n}{p}} \|X\|_\infty. \quad (6.15)$$

Step 2: passing to Frobenius norm. We have not used the low rank assumption yet, and will do this now. Since $\text{rank}(X) \leq r$ by assumption and $\text{rank}(\hat{X}) \leq r$ by construction, we have $\text{rank}(\hat{X} - X) \leq 2r$. The relationship (4.4) between the operator and Frobenius norms thus gives

$$\|\hat{X} - X\|_F \leq \sqrt{2r} \|\hat{X} - X\|.$$

Taking expectations and using the bound on the error in the operator norm (6.15), we get

$$\mathbb{E} \|\hat{X} - X\|_F \leq \sqrt{2r} \mathbb{E} \|\hat{X} - X\| \lesssim \sqrt{\frac{rn \log n}{p}} \|X\|_\infty.$$

Dividing both sides by n , we can rewrite this bound as

$$\mathbb{E} \frac{1}{n} \|\hat{X} - X\|_F \lesssim \sqrt{\frac{rn \log n}{pn^2}} \|X\|_\infty.$$

To finish the proof, recall that $pn^2 = m$ by the definition (6.12) of p . □

Exercise 6.6.2 (Bounding rows of random matrices). 🍷 Consider i.i.d. random variables $\delta_{ij} \sim \text{Ber}(p)$, where $i, j = 1, \dots, n$. Assuming that $pn \geq \log n$, show that

$$\mathbb{E} \max_{i \in [n]} \sum_{j=1}^n (\delta_{ij} - p)^2 \leq Cpn.$$

Hint: Fix i and use Bernstein's inequality to get a tail bound for $\sum_{j=1}^n (\delta_{ij} - p)^2$. Conclude by taking a union bound over $i \in [n]$.

Exercise 6.6.3 (Rectangular matrices). 🍷 State and prove a version of the Matrix Completion Theorem 6.6.1 for general rectangular $n_1 \times n_2$ matrices X .

Exercise 6.6.4 (Noisy observations). 🍷 Extend the Matrix Completion Theorem 6.6.1 to noisy observations, where we are shown noisy versions $X_{ij} + \nu_{ij}$ of

some entries of X . Here ν_{ij} are independent and mean zero sub-gaussian random variables representing noise.

Remark 6.6.5 (Improvements). The logarithmic factor can be removed from the bound of Theorem 6.6.1, and in some cases matrix completion can be *exact*, i.e. with zero error. See the notes after this chapter for details.

6.7 Contraction Principle

We conclude this chapter with one more useful inequality. We keep denoting by $\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots$ a sequence of independent symmetric Bernoulli random variables (which is also independent of any other random variables in question).

Theorem 6.7.1 (Contraction principle). *Let $x_1, \dots, x_N$ be (deterministic) vectors in some normed space, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Then*

$$\mathbb{E} \left\| \sum_{i=1}^N a_i \varepsilon_i x_i \right\| \leq \|a\|_\infty \cdot \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i x_i \right\|.$$

Proof Without loss of generality, we may assume that $\|a\|_\infty \leq 1$. (Why?) Define the function

$$f(a) := \mathbb{E} \left\| \sum_{i=1}^N a_i \varepsilon_i x_i \right\|. \quad (6.16)$$

Then $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is a convex function. (See Exercise 6.7.2.)

Our goal is to find a bound for f on the set of points a satisfying $\|a\|_\infty \leq 1$, i.e. on the unit cube $[-1, 1]^n$. By the elementary maximum principle for convex functions, the maximum of a convex function on a compact set in $\mathbb{R}^n$ is attained at an extreme point of the set. Thus f attains its maximum at one of the vertices of the cube, i.e. at a point a whose coefficients are all $a_i = \pm 1$.

For this point a , the random variables $(\varepsilon_i a_i)$ have the same distribution as (ε_i) due to symmetry. Thus

$$\mathbb{E} \left\| \sum_{i=1}^N a_i \varepsilon_i x_i \right\| = \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i x_i \right\|,$$

Summarizing, we showed that $f(a) \leq \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i x_i \right\|$ whenever $\|a\|_\infty \leq 1$. This completes the proof. $\square$

Exercise 6.7.2.  Check that the function f defined in (6.16) is convex.

Exercise 6.7.3 (Contraction principle for general distributions).  Prove the following generalization of Theorem 6.7.1. Let $X_1, \dots, X_N$ be independent, mean zero random vectors in a normed space, and let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. Then

$$\mathbb{E} \left\| \sum_{i=1}^N a_i X_i \right\| \leq 4 \|a\|_\infty \cdot \mathbb{E} \left\| \sum_{i=1}^N X_i \right\|.$$

Hint: Use symmetrization, contraction principle (Theorem 6.7.1) conditioned on (X_i) , and finish by applying symmetrization again.

As an application, let us show how symmetrization can be done using *Gaussian* random variables $g_i \sim N(0, 1)$ instead of symmetric Bernoulli random variables ε_i .

Lemma 6.7.4 (Symmetrization with Gaussians). *Let $X_1, \dots, X_N$ be independent, mean zero random vectors in a normed space. Let $g_1, \dots, g_N \sim N(0, 1)$ be independent Gaussian random variables, which are also independent of X_i . Then*

$$\frac{c}{\sqrt{\log N}} \mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 3 \mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\|.$$

Proof **Upper bound.** By symmetrization (Lemma 6.4.2), we have

$$E := \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|.$$

To interject Gaussian random variables, recall that $\mathbb{E} |g_i| = \sqrt{2/\pi}$. Thus we can continue our bound as follows:³

$$\begin{aligned} E &\leq 2\sqrt{\frac{\pi}{2}} \mathbb{E}_X \left\| \sum_{i=1}^N \varepsilon_i \mathbb{E}_g |g_i| X_i \right\| \\ &\leq 2\sqrt{\frac{\pi}{2}} \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i |g_i| X_i \right\| \quad (\text{by Jensen's inequality}) \\ &= 2\sqrt{\frac{\pi}{2}} \mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\|. \end{aligned}$$

The last equality follows by symmetry of Gaussian distribution, which implies that the random variables $\varepsilon_i |g_i|$ have the same distribution as g_i (recall Exercise 6.4.1).

Lower bound can be proved by using contraction principle (Theorem 6.7.1)

³ Here we use index g in $\mathbb{E}_g$ to indicate that this is an expectation “over (g_i) ”, i.e. conditional on (X_i) . Similarly, E_X denotes the expectation over (X_i) .

and symmetrization (Lemma 6.4.2). We have

$$\begin{aligned}
\mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\| &= \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i g_i X_i \right\| \quad (\text{by symmetry of } g_i) \\
&\leq \mathbb{E}_g \mathbb{E}_X \left(\|g\|_\infty \mathbb{E}_\varepsilon \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right) \quad (\text{by Theorem 6.7.1}) \\
&= \mathbb{E}_g \left(\|g\|_\infty \mathbb{E}_\varepsilon \mathbb{E}_X \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right) \quad (\text{by independence}) \\
&\leq 2 \mathbb{E}_g \left(\|g\|_\infty \mathbb{E}_X \left\| \sum_{i=1}^N X_i \right\| \right) \quad (\text{by Lemma 6.4.2}) \\
&= 2 \left(\mathbb{E} \|g\|_\infty \right) \left(\mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \right) \quad (\text{by independence}).
\end{aligned}$$

It remains to recall from Exercise 2.5.10 that

$$\mathbb{E} \|g\|_\infty \leq C \sqrt{\log N}.$$

The proof is complete. $\square$

Exercise 6.7.5. $\clubsuit\clubsuit$ Show that the factor $\sqrt{\log N}$ in Lemma 6.7.4 is needed in general, and is optimal. Thus, symmetrization with Gaussian random variables is generally weaker than symmetrization with symmetric Bernoullis.)

Exercise 6.7.6 (Symmetrization and contraction for functions of norms). Let $F : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a convex increasing function. Generalize the symmetrization and contraction results of this and previous section by replacing the norm $\|\cdot\|$ with $F(\|\cdot\|)$ throughout.

In the following exercise we set foot in the study of random processes, which we fully focus on in the next chapter.

Exercise 6.7.7 (Talagrand's contraction principle). $\clubsuit\clubsuit\clubsuit$ Consider a bounded subset $T \subset \mathbb{R}^n$, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent symmetric Bernoulli random variables. Let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be contractions, i.e. Lipschitz functions with $\|\phi_i\|_{\text{Lip}} \leq 1$. Then

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^n \varepsilon_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^n \varepsilon_i t_i. \quad (6.17)$$

To prove this result, do the following steps:

- (a) First let $n = 2$. Consider a subset $T \subset \mathbb{R}^2$ and contraction $\phi : \mathbb{R} \rightarrow \mathbb{R}$, and check that

$$\sup_{t \in T} (t_1 + \phi(t_2)) + \sup_{t \in T} (t_1 - \phi(t_2)) \leq \sup_{t \in T} (t_1 + t_2) + \sup_{t \in T} (t_1 - t_2).$$

(b) Use induction on n complete proof.

Hint: To prove (6.17), condition on $\varepsilon_1, \dots, \varepsilon_{n-1}$ and apply part 1.

Exercise 6.7.8. 🍷 Generalize Talagrand’s contraction principle for arbitrary Lipschitz functions $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ without restriction on their Lipschitz norms.

Hint: Theorem 6.7.1 may help.

6.8 Notes

A version of the decoupling inequality we stated in Theorem 6.1.1 and Exercise 6.1.5 was originally proved by J. Bourgain and L. Tzafriri [32]. We refer the reader to the papers [61] and books [62], [78, Section 8.4] for related results and extensions.

The original form of Hanson-Wright inequality, which is somewhat weaker than Theorem 6.2.1, goes back to [97, 228]. The version of Theorem 6.2.1 and its proof we gave in Section 6.2 are from [179]. Several special cases of Hanson-Wright inequality appeared earlier in [78, Proposition 8.13] for Bernoulli random variables, in [199, Lemma 2.5.1] for Gaussian random variables, and in [17] for diagonal-free matrices.

Concentration for anisotropic random vectors (Theorem 6.3.2) and the bound on the distance between a random vectors and a subspace (Exercise 6.3.4) are taken from [179].

Symmetrization Lemma 6.4.2 and its proof can be found e.g. in [130, Lemma 6.3], [78, Section 8.2].

Although the precise statement of Theorem 6.5.1 is difficult to locate in existing literature, the result is essentially known. It can be deduced, for example, from the inequalities in [206, 207]. The factor $\sqrt{\log n}$ in Theorem 6.5.1 can be improved to $\log^{1/4} n$ by combining a result of Y. Seginer [183, Theorem 3.1] with symmetrization (Lemma 6.4.2); see [15, Corollary 4.7] for an alternative approach to Seginer’s theorem. This improved factor is optimal as is demonstrated by the result of Exercise 6.5.4, which is due to Y. Seginer [183, Theorem 3.2]. Moreover, for many classes of matrices the factor $\log^{1/4} n$ can be removed completely; this happens, in particular, for matrices with i.i.d. entries [183] and matrices with Gaussian entries [125]. We refer the reader to [213, Section 4], [125] for more elaborate results, which describe the operator norm of the random matrix A in terms of the variances of its entries.

Theorem 6.6.1 on matrix completion and its proof are from [170, Section 2.5], although versions of it may have appeared before. In particular, Keshavan, Montanari and Oh [113] showed how to obtain a slightly better bound—one without the logarithmic factor—by “trimming” the random matrix Y , where one removes the rows and columns of Y that have, say, twice more many nonzero entries than expected. E. Candes and B. Recht [45] demonstrated that under some additional incoherence assumptions, *exact* matrix completion (with zero error) is possible with $m \asymp rn \log^2(n)$ randomly sampled entries. We refer the reader to the papers [47, 173, 92, 57] for many further developments on matrix completion.

The contraction principle (Theorem 6.7.1) is taken from [130, Section 4.2]; see also [130, Corollary 3.17, Theorem 4.12] for different versions of the contraction principle for random processes. Lemma 6.7.4 can be found in [130, inequality (4.9)]. While the logarithmic factor is in general needed there, it can be removed if the normed space has non-trivial cotype, see [130, Proposition 9.14]. Talagrand's contraction principle (Exercise 6.7.7) can be found in [130, Corollary 3.17], where one can find a more general result (with a convex and increasing function of the supremum). Exercise 6.7.7 is adapted from [212, Exercise 7.4]. A Gaussian version of Talagrand's contraction principle will be given in Exercise 7.2.13.